Stockholm University

Department of Meteorology

Numerical Methods in Atmospheric Sciences and Oceanography (MO8007)

Lab 2 – The Shallow Water Equation

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1 Introduction

Numerical weather models ... Döös et al. [2022].

The shallow water equations are a simplified version of the Navier-Stokes equations, which govern fluid flow. The shallow water equations are

$$\begin{cases} \dfrac{\partial u}{\partial t} + g\dfrac{\partial h}{\partial x} = 0 & \text{(1a)} \\ \dfrac{\partial h}{\partial t} + H\dfrac{\partial u}{\partial x} = 0 & \text{(1b)} \end{cases}$$

The analytical solution to the shallow water equations, assuming periodic boundary conditions and initial velocity of $u(x,0) = 0$, can be written as two opposing travelling waves

$$h(x,t) = h_1(x - c\,t) + h_2(x + c\,t),$$
$$u(x,t) = u_1(x - c\,t) + u_2(x + c\,t),$$

where $c = \sqrt{gH}$ is the phase speed of the waves. The initial condition $h(x,t) = h_0(x)$ and $u(x,0) = 0$ can be used together with Equation 1 to find h_1, h_2, u_1 and u_2. From the initial condition, we can find that $h_1(x) + h_2(x) = h_0(x)$, and from the initial velocity we can find that $u_1(x) + u_2(x) = 0$. Using Equation 1 one can derive that $u_1(x) = \frac{c}{H}h_1(x)$ and $u_2(x) = -\frac{c}{H}h_2(x)$. This gives the following solution for the shallow water equations:

$$h(x,t) = \frac{1}{2}h_0(x - c\,t) + \frac{1}{2}h_0(x + c\,t), \tag{2a}$$
$$u(x,t) = \frac{c}{2H}h_0(x - c\,t) - \frac{c}{2H}h_0(x + c\,t). \tag{2b}$$

1.1 The numerical models

1.1.1 Leapfrog scheme on an unstaggered grid

Using the leapfrog scheme for an unstaggered grid, the shallow water equations (Equation 1) can be discretized as:

$$\frac{u_j^{n+1} - u_j^{n-1}}{2\Delta t} = -g\frac{h_{j+1}^n - h_{j-1}^n}{2\Delta x} \tag{3a}$$
$$\frac{h_j^{n+1} - h_j^{n-1}}{2\Delta t} = -H\frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x}, \tag{3b}$$

where the grid was defined as $x_j = j\Delta x$ and $t^n = n\Delta t$. And the courant number μ is defined as $\mu = \frac{c\,\Delta t}{\Delta x}$. This gives the following step equations for u and h:

$$u_j^{n+1} = u_j^{n-1} - \frac{g\Delta t}{\Delta x}\left(h_{j+1}^n - h_{j-1}^n\right) \tag{4a}$$
$$h_j^{n+1} = h_j^{n-1} - \frac{H\Delta t}{\Delta x}\left(u_{j+1}^n - u_{j-1}^n\right) \tag{4b}$$

One can note that this is stable if $\mu \leq 1$.

However, this scheme uses the fact that we have the values of u and h at time steps $n-1$ and n, which means that we need to initialize the model somehow. A common way of doing this is using a forward Euler step for the first time step, which gives:

$$u_j^1 = u_j^0 - \frac{g\Delta t}{2\Delta x}\left(h_{j+1}^0 - h_{j-1}^0\right) \tag{5a}$$
$$h_j^1 = h_j^0 - \frac{H\Delta t}{2\Delta x}\left(u_{j+1}^0 - u_{j-1}^0\right) \tag{5b}$$

1.1.2 Leapfrog scheme on an staggered grid

If one instead uses a leapfrog scheme with a staggered grid, the shallow water equations (Equation 1) can be discretized as:

$$\frac{u_j^{n+1} - u_j^{n-1}}{2\Delta t} = -g\frac{h_{j+1}^n - h_j^n}{\Delta x} \tag{6}$$

$$\frac{h_j^{n+1} - h_j^{n-1}}{2\Delta t} = -H\frac{u_{j+1}^n - u_j^n}{\Delta x}, \tag{7}$$

where the grid now uses $x_j = j\Delta x$ for h_j and $x_{j+\frac{1}{2}} = (j + \frac{1}{2})\Delta x$ for u_j (if one uses the same j-coordinate for u as in h). This gives the following step equations for u and h:

$$u_j^{n+1} = u_j^{n-1} - \frac{2g\Delta t}{\Delta x}\left(h_{j+1}^n - h_j^n\right) \tag{8a}$$

$$h_j^{n+1} = h_j^{n-1} - \frac{2H\Delta t}{\Delta x}\left(u_{j+1}^n - u_j^n\right) \tag{8b}$$

This scheme is stable if $\mu \leq 0.5$. However, it also needs to be initialized, and the same forward Euler step method can be used for this scheme.

1.1.3 Robert-Asselin filtering

There are different ways to get rid of this computational mode and thus get cleaner results. One common way is to use some kind of filter, such as the Robert-Asselin filter. The Robert-Asselin filter works by smoothing the result, since the computational mode adds background noise. This is done as

$$u_{\text{filtered}}^n = u^n + \gamma\left(u_{\text{filtered}}^{n-1} - 2u^n + u^{n+1}\right), \tag{9}$$

where γ is a filtering parameter. The Robert-Asselin filter has the drawback in that a it also dampens the result by a factor $[1 - 4\gamma\sin(\omega\Delta t/2)]$ due to the dispersion-like smoothing. Furthermore, this parameter also affects the stability criteria so that, for example, in the unstaggared Leapfrog scheme the stability condition becomes $\mu + \gamma \leq 1$. For the staggered Leapfrog scheme, the stability condition becomes $\mu + \gamma \leq 0.5$.

1.2 Error computation

The error between a numerical scheme and analytical solution can be quantified using the norm. For a vector $\mathbf{h}^n$, the ℓ_2-norm is defined as

$$\|\mathbf{h}^n\| = \sqrt{\frac{1}{N}\sum_{j=1}^{N}(h_j^n)^2}. \tag{10}$$

To obtain a dimensionless measure of the error, the relative error can be computed at time step n as

$$\text{error}^n = \frac{\|\mathbf{h}_{\text{numerical}}^n - \mathbf{h}_{\text{analytical}}^n\|}{\|\mathbf{h}_{\text{analytical}}^n\|}. \tag{11}$$

2 Methodology

The numerical solution of the shallow water equations was implemented using two different grids: an unstaggered grid and a staggered grid. The initial condition for the height field h was defined as a

localized cosine wave perturbation given by

$$h_0(x) = \begin{cases} 0.5 + 0.5\cos\left(10\pi(x-0.5)\right), & 0.4 < x < 0.6, \\ 0, & \text{otherwise.} \end{cases} \tag{12}$$

And a velocity field u initialized as zero. Both numerical schemes were implemented using the leapfrog time-stepping method, for a total simulation time of $T = 1$ arbitrary time units and a spatial domain of $x \in [0, 1]$ arbitrary length units, with periodic boundary conditions. Both models used a phase speed of $c = 1$, mean fluid depth of $H = 1$ and a gravitational acceleration of $g = 1$.

The unstaggered grid scheme was implemented with a spatial resolution of $\Delta x = 0.025$ and a a Courant number of $\mu = 0.9$. The staggered grid scheme was implemented with three different spatial resolutions: $\Delta x = 0.05$, $\Delta x = 0.025$ and $\Delta x = 0.0125$, all with a Courant number of $\mu = 0.45$.

3 Results

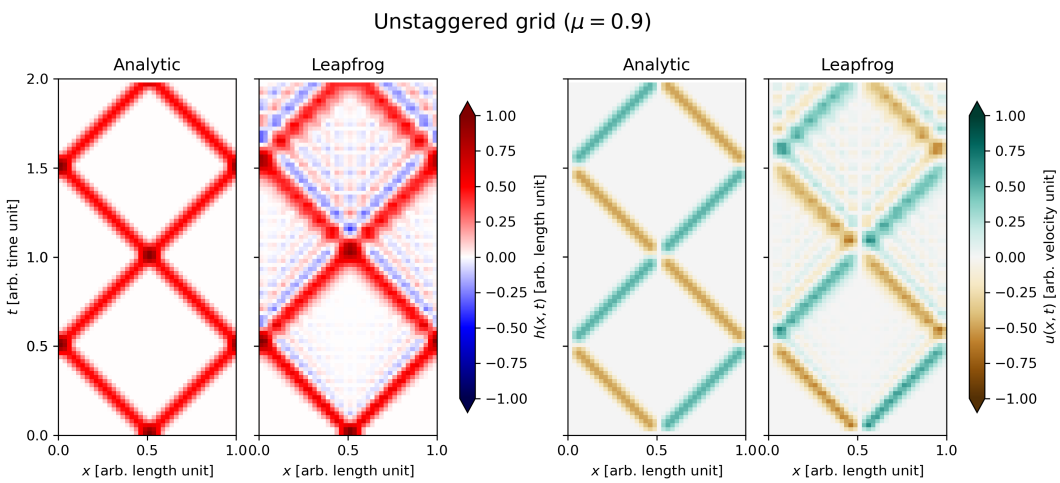

Figure 1: Hovmöller diagram of the numerical solution using the unstaggered grid. The propagation of the wave $h(x, t)$ and its velocity $u(x, t)$ is shown over time.

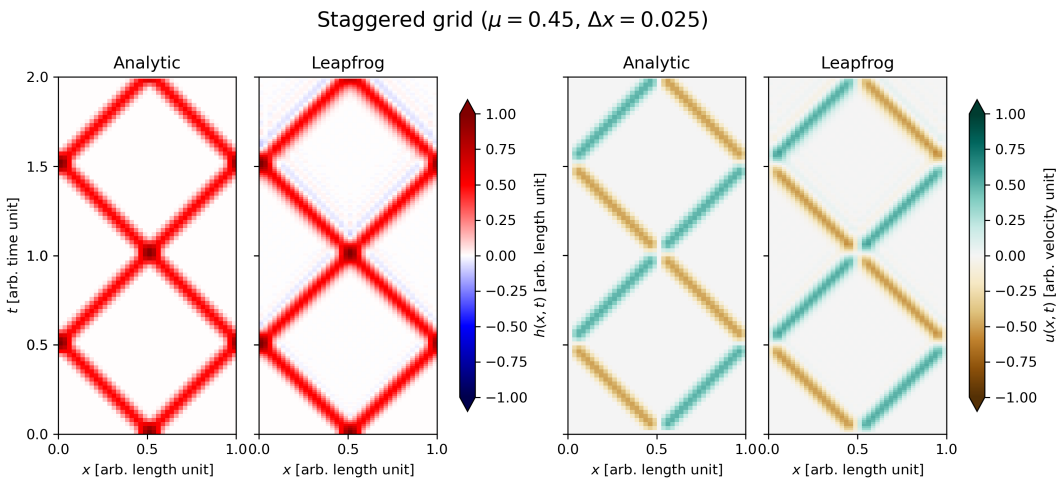

Figure 2: Hovmöller diagram of the numerical solution using the staggered grid. The propagation of the wave $h(x, t)$ and its velocity $u(x, t)$ is shown over time.

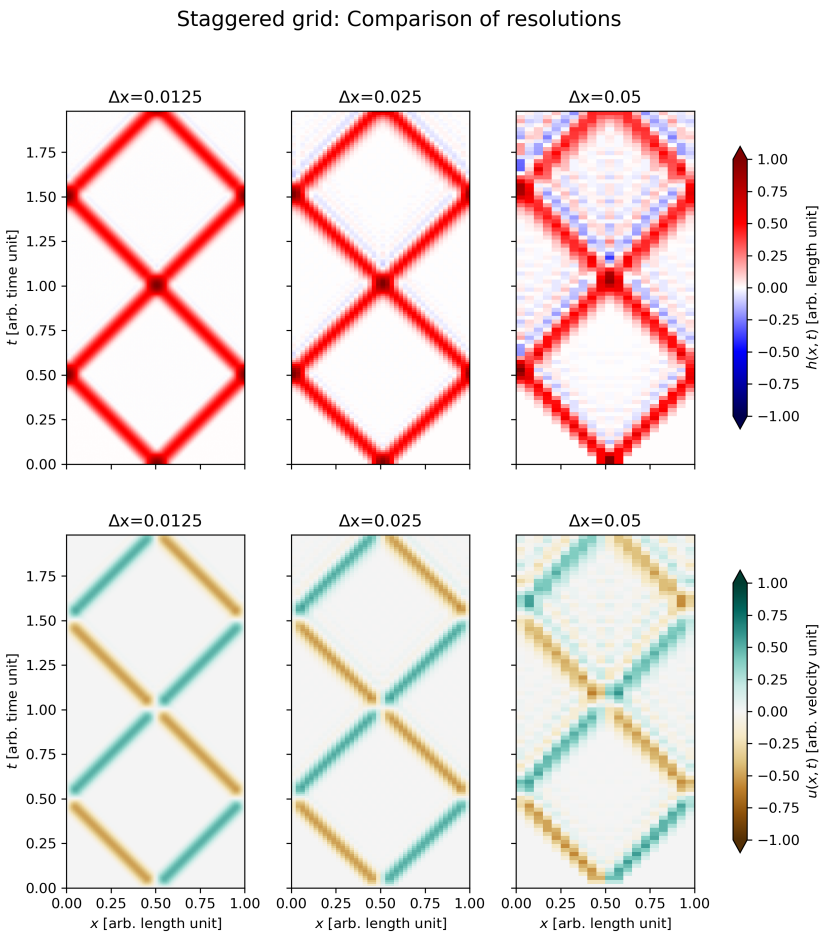

Figure 3: Comparison of staggered grid solutions at different spatial resolutions.

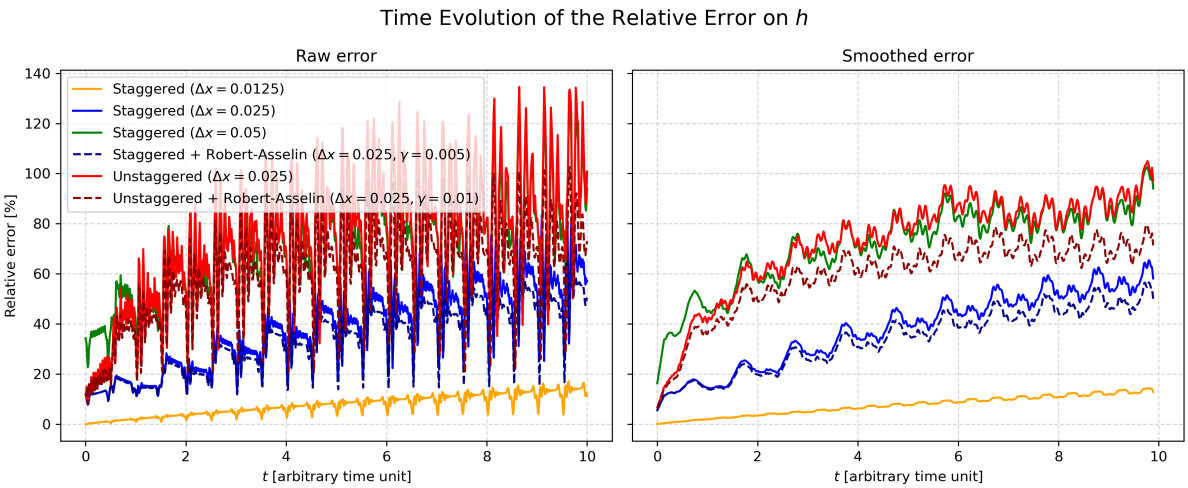

Figure 4: Time evolution of the relative error in h for different numerical schemes. The smoothed error curves uses a moving average filter and are shown for clearer visualization.

4 Discussion

5 Conclusion

References

Kristofer Döös, Peter Lundberg, and Aitor Aldama Campino. *Basic Numerical Methods in Meteorology and Oceanography*. Stockholm University Press, Stockholm, Sweden, 2022. ISBN 978-91-7635-172-7. doi: 10.16993/bbs.

A Acknowledgements

The text in this report was spell-checked using Mistral AI's Le Chat. The schemes in this report were obtained using the Fortan programming language, while the figures were created using the Python programming language.